

Assignment 8: Time Series Analysis

Cira Ortiz

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#Set up
library(here)
```

```
## here() starts at /home/guest/ciraortizenviron872
```

```
here()
```

```
## [1] "/home/guest/ciraortizenviron872"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.2
## v ggplot2    4.0.0      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

```
library(zoo)
```

```
##
```

```
## Attaching package: 'zoo'
```

```
##
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## as.Date, as.Date.numeric
```

```
library(trend)
```

```
halloweentHEME<-mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "bottom",
        legend.background =element_rect(fill ="orange"),
        plot.title = element_text(hjust = 0.5),
        plot.background = element_rect(fill = "orange"))
```

```
theme_set(halloweentHEME)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
```

```
data2010<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2011<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2012<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2013<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2014<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2015<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2016<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
  stringsAsFactors = TRUE)
```

```
data2017<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
  stringsAsFactors = TRUE)

data2018<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
  stringsAsFactors = TRUE)

data2019<-read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
  stringsAsFactors = TRUE)

GaringerOzone<-bind_rows(data2010,data2011,data2012,data2013,data2014,data2015,data2016,
  data2017,data2018,data2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily data set, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3 - setting date as date class
GaringerOzone$Date<- as.Date(as.character(GaringerOzone$Date), format = "%m/%d/%Y")

# 4 - only Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE
selectcol<-select(GaringerOzone,
  Date,
  Daily.Max.8.hour.Ozone.Concentration,
  DAILY_AQI_VALUE)

# 5 - filling in NAs
Days<-as.data.frame(seq
  (from =as.Date("2010-01-01"),
  to = as.Date("2019-12-31"),
  by = "day"))
colnames(Days)<-"Date"

# 6 - Combining dataframes using left join
GaringerOzone<-left_join(Days, selectcol)
```

```
## Joining with 'by = join_by(Date)'
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear

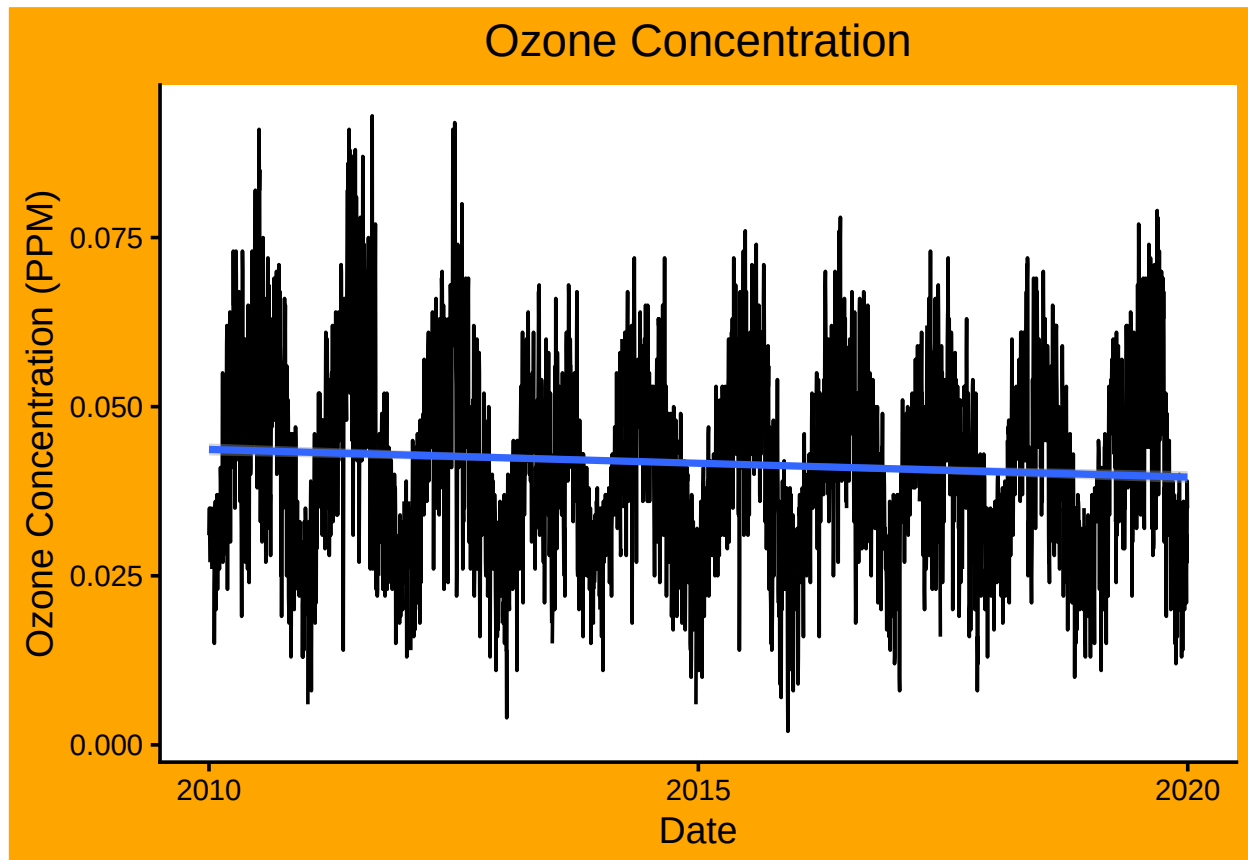
trend of your data. Does your plot suggest a trend in ozone concentration over time?

#7 - Line plot for ozone concentrations over time + smoothed line

```
ex7plot<-GaringerOzone %>%  
  ggplot(aes(x =Date,  
             y = Daily.Max.8.hour.Ozone.Concentration))+  
  geom_line()+  
  geom_smooth(method = "lm")+  
  labs(x = "Date", y = "Ozone Concentration (PPM)",  
       title = "Ozone Concentration")  
  
ex7plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```



Answer: The plot suggests a decreasing ozone concentration over from 2010 to 2020, still within the range of 0.050 and 0.025.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 - Linear interpolation for missing daily data
#identifying NAs = (63)
head(GaringerOzone)
```

```
##           Date Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## 1 2010-01-01                      0.031                      29
## 2 2010-01-02                      0.033                      31
## 3 2010-01-03                      0.035                      32
## 4 2010-01-04                      0.031                      29
## 5 2010-01-05                      0.027                      25
## 6 2010-01-06                      NA                       NA
```

```
summary(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration)
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.      NA's
## 0.00200 0.03200 0.04100 0.04163 0.05100 0.09300         63
```

```
# Adding new column with no missing obs
GaringerOzone.clean <-
  GaringerOzone %>%
  mutate(Ozone.clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration) )

#Checking to see if NAs are removed
summary(GaringerOzone.clean$Ozone.clean)
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: Since we're looking at daily ozone concentrations over several years, a linear interpolation is the best option since it fills in NAs with values between the previous and next measurement. This accounts for the daily variability in ozone levels, whereas a piecewise constant could create a no change segments in the trend that is not accurate since it's assuming the NAs value are equal to the nearest neighbor. Similarly, a spline could introduce inaccurate peaks and trends that also aren't representative.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9 - New data frame of aggregated data
GaringerOzone.monthly<- GaringerOzone.clean %>%
  mutate(
    Year = year(Date),
    Month = month(Date)) %>%
  group_by(Year, Month) %>%
  summarise(Mean_Ozone = mean(Ozone.clean),
    .groups="keep") #added to keep groupings
```

```
#creating a new date column
GaringerOzone.monthly<-GaringerOzone.monthly %>%
  mutate(Date=my(paste0(Month,"-",Year)))
```

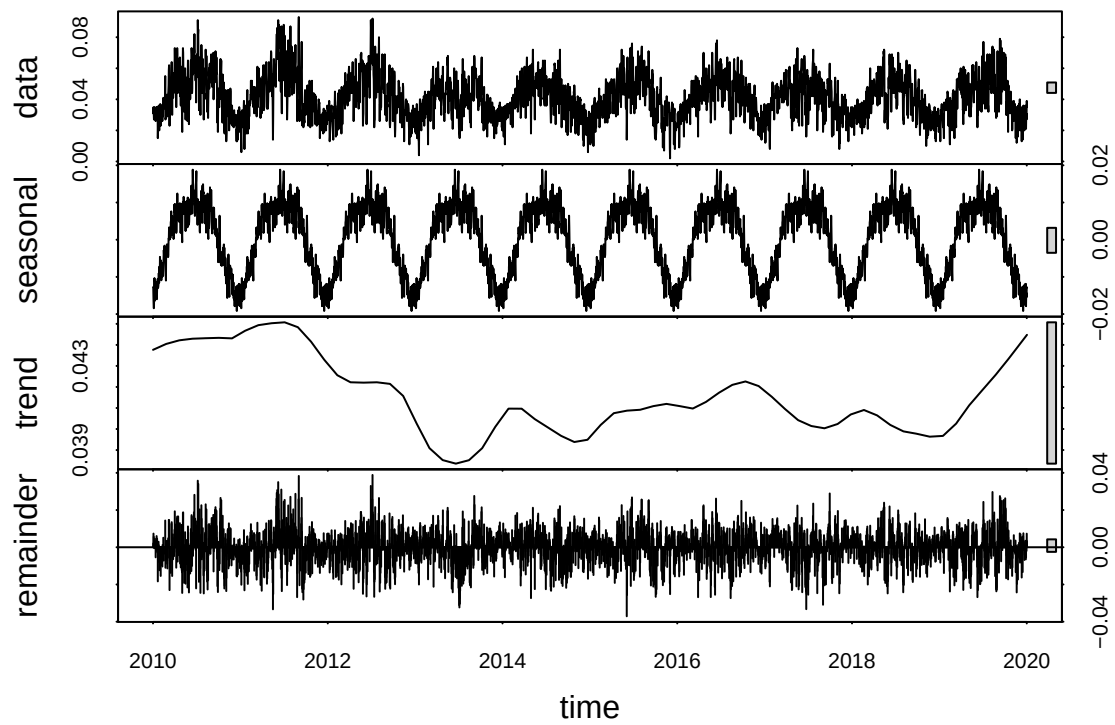
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10 - Time series object for daily data
#defining time series object, daily data increased by 1 day every time it runs
GaringerOzone.daily.ts <- ts(GaringerOzone.clean$Ozone.clean, start = c(2010,1), frequency = 365)
#Time series object for monthly data
GaringerOzone.monthly.ts<-ts(GaringerOzone.monthly$Mean_Ozone, start = c(2010,1), frequency = 12)
```

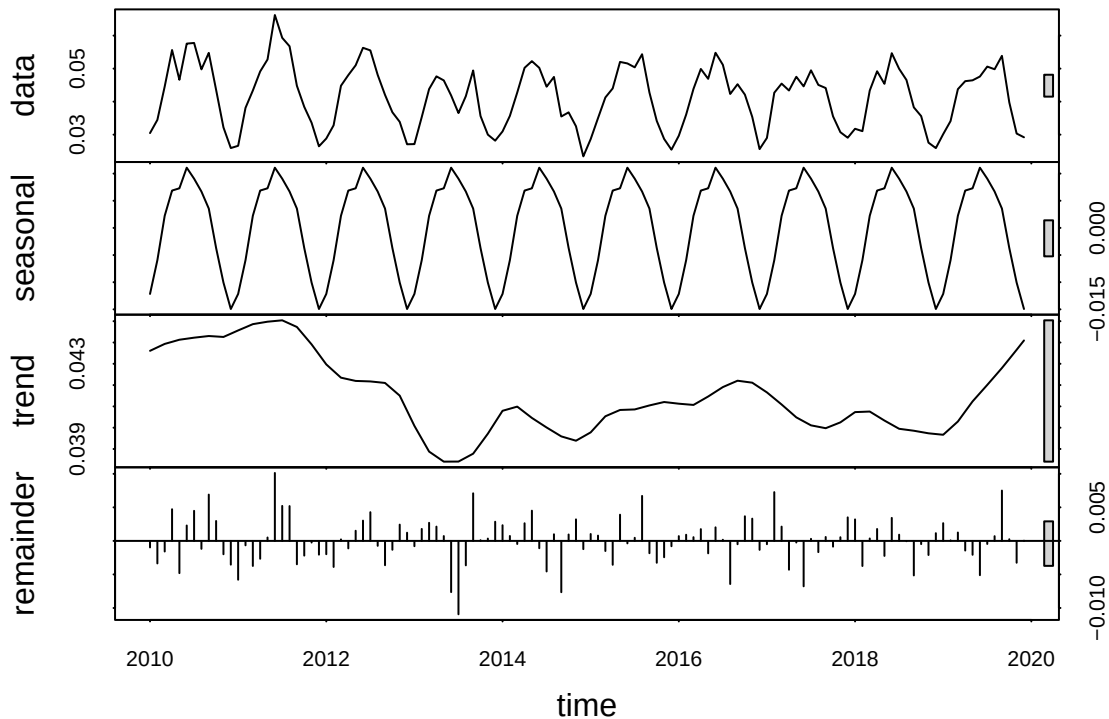
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11 - Decompose daily data
daily.decomposed<-stl(GaringerOzone.daily.ts, s.window ="periodic")
#Monthly data decomposed
monthly.decomposed<-stl(GaringerOzone.monthly.ts, s.window = "periodic")

#Ploting decomposed data
plot(daily.decomposed)
```



```
plot(monthly.decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
# Run SMK test
smk.monthly.ozone <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

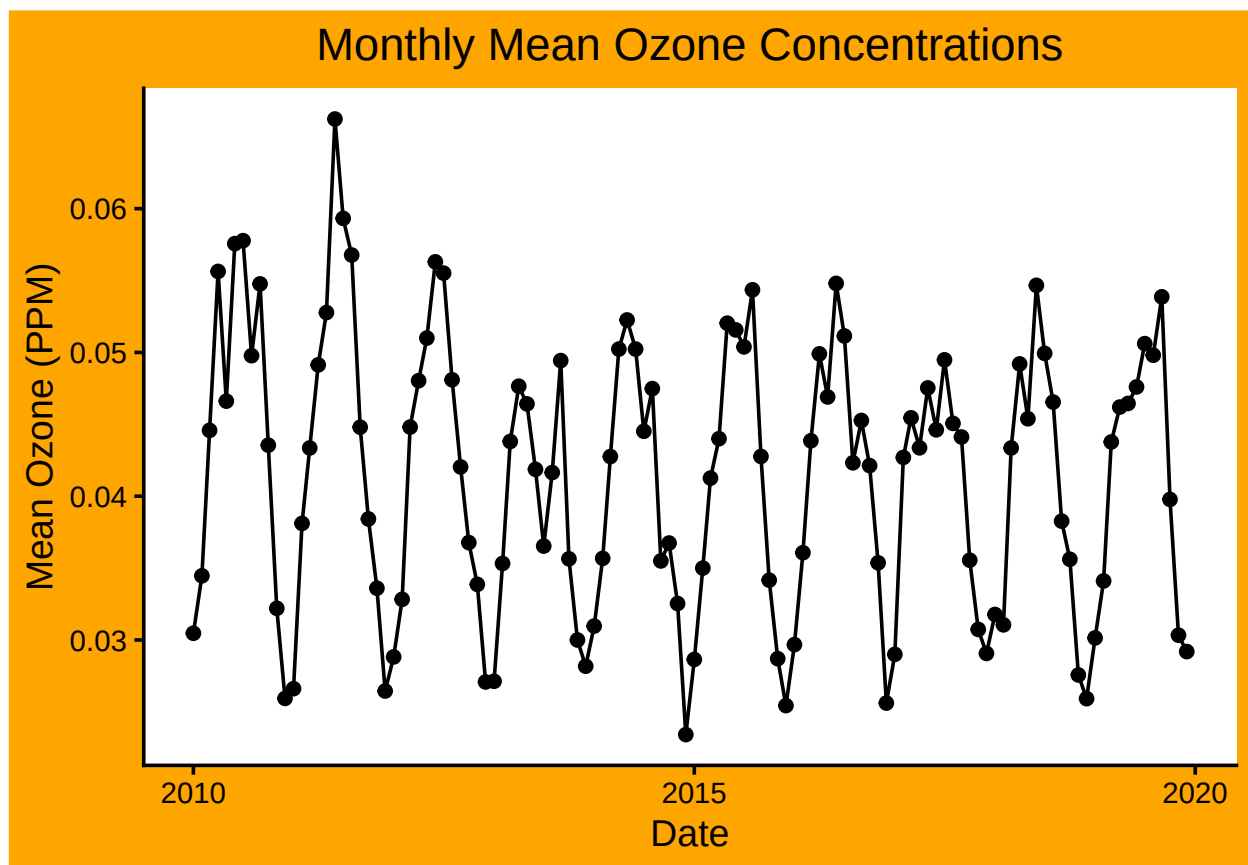
# Inspect results
summary(smk.monthly.ozone)

## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: The seasonal version of the mann-kendall test is most appropriate since it accounts for the seasonable variability (due to temperature, sunlight, etc.) that we can see in the ozone monthly data.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.


```
# 13 - Plot of mean monthly ozone concentration
monthly.ozone<-GaringerOzone.monthly %>%
  ggplot(aes(
    x=Date,
    y=Mean_Ozone))+
  geom_point()+
  geom_line()+
  labs(x="Date",
       y="Mean Ozone (PPM)",
       title="Monthly Mean Ozone Concentrations")
print(monthly.ozone)
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation. Have ozone concentrations changed over the 2010s at this station?

Answer: When looking at the decomposed data, it is evident that much of the variability in the data is explained by seasonality. The trend line (seen on the decomposed data) reflects a small dip around 2013, but for the most part the trend is fairly stable, with an upward slope headed into 2020. Overall however, the station's ozone concentration's have not changed significantly.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15- Extract seasonality
```

```
noszn.ozone.components <- as.data.frame(monthly.decomposed$time.series[,2:3]) #removed 1 (the seasonal i
```

```
#16- Mann Kendall test
```

```
noszn.ozone.components.trend <- Kendall::MannKendall(GaringerOzone.monthly.ts)
```

```
summary(noszn.ozone.components.trend)
```

```
## Score = -424 , Var(Score) = 194364.7
```

```
## denominator = 7139
```

```
## tau = -0.0594, 2-sided pvalue =0.33732
```

```
summary(smk.monthly.ozone)
```

```
## Score = -77 , Var(Score) = 1499
```

```
## denominator = 539.4972
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: After removing seasonality the tau value was still negative indicating a downward trend, but the p value is still greater than .05, meaning it's not a statistically significant trend between time (month) and ozone concentration. In comparison, the seasonal mann kendall test had a p-value of .0467 indicating a statistically significant trend however the trend was a small decline represented by the S score.